Feature-Level vs Sample-Level Weights

Category-shift cold-start on text multiclass
(FULL LIVE holdout; no EWMA)

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Abstract

Both weight channels share one FSDS RF-domain attribution (LIVE vs REF ⇒ VIMP), then act on *orthogonal axes*: **feature-level** rescales coordinates ($x_{ij} \leftarrow x_{ij}\, w_j^{\text{feat}}$); **sample-level** rescales CE mass (`partial_fit(..., sample_weight=`w_i`)`). On text boards, mean Δoverall-acc vs uniform is +0.009 (feature), +0.021 (sample), +0.023 (both); mean novel OVA-AUC is essentially flat.

1 Shared upstream: RF-domain VIMP

Let $X^{\text{ref}} \in \mathbb{R}^{n_0 \times p}$ be the known-category reference and $X^{\text{live}} \in \mathbb{R}^{n_1 \times p}$ the live batch (known + novel). Fit a RF on

$$W_i = \begin{cases} 0 & i \in \text{REF}, \\ 1 & i \in \text{LIVE}, \end{cases}$$

and take feature importances

$$v \in \Delta^{p-1}, \qquad v_j = \frac{\text{Imp}_j}{\sum_k \text{Imp}_k}.$$

Coordinates with large v_j are those that best separate the live batch from the reference—the domain-shift footprint under category arrival.

2 Feature-level weights (geometry)

Normalize relative to the top attributed coordinate and stretch:

$$\tilde{v}_j = \frac{v_j}{\max_k v_k}, \qquad w_j^{\text{feat}} = 1 + \alpha\, \tilde{v}_j \qquad (\alpha = 3 \text{ default}). \tag{1}$$

Apply the same scale to REF and LIVE,

$$x_{ij} \;\leftarrow\; x_{ij}\, w_j^{\text{feat}}, \tag{2}$$

then re-init the CE head on scaled REF so coefficients live in the new geometry. Top VIMP coords get $\approx 1 + \alpha = 4\times$; quiet coords stay near 1.

Role. Puts CE gradient budget on the axes that carry the category shift. Especially natural for TF-IDF / BoW: a few token dimensions encode the new category; most coordinates are noise. Feature-level does *not* drop samples—it reweights the representation.

3 Sample-level weights (mass)

3.1 Membership in the attributed subspace

Take the top-K VIMP indices $\mathcal{S} = \arg\operatorname{top}_K(v)$. On $\mathcal{S}$, z-score LIVE against REF mean/std and set

$$d_i = \|z_i^{(\mathcal{S})}\|_2, \qquad m_i = \operatorname{clip}\Big(\frac{d_i - q_{0.2}(d)}{q_{0.8}(d) - q_{0.2}(d)}, 0, 1\Big). \tag{3}$$

High m_i = far from REF in the shift subspace = "new-category-like". Optional private-live fuse (labels available at adapt time):

$$m_i \;\leftarrow\; \max(m_i,\, \mathbf{1}[y_i \in \mathcal{C}_{\text{novel}}]). \tag{4}$$

3.2 Soft + hard → fused

$$w_i^{\text{soft}} = 1 + \lambda_{\text{eff}}\, m_i, \tag{5}$$

$$w_i^{\text{subset}} = \begin{cases} \beta_{\text{eff}} & m_i \geq q, \\ 1 & \text{else}, \end{cases} \tag{6}$$

$$w_i = \frac{n\,(w^{\text{soft}} \odot w^{\text{subset}})_i}{\sum_{i'}(w^{\text{soft}} \odot w^{\text{subset}})_{i'}}. \tag{7}$$

$\lambda_{\text{eff}}, \beta_{\text{eff}}$ grow with membership concentration (sharp footprint ⇒ stronger boost). Pass w into `partial_fit(`X`, `y`, sample_weight=`w`)`.

Role. Raises the CE share of novel / high-membership points without discarding known-class rows. Orthogonal to (1): same geometry, different row masses.

4 Four schemes

Scheme	Feature axis	Sample axis
`uniform`	raw X	$w_i \equiv 1$
`feature_w`	$X \cdot w^{\text{feat}}$ (1)	$w_i \equiv 1$
`sample_w`	raw X	fused w (7)
`both`	$X \cdot w^{\text{feat}}$	fused w

Loss = CE (`log_loss`) SGD; no EWMA. KPI = overall accuracy + mean novel OVA-AUC on a *fixed full LIVE holdout* (not the first-n arrival window). Curves vary novel adapt budget $b \in \{20, 40, \ldots, 200\}$ with known context $\approx 2b$.

5 Why two levers

- **Feature-level** answers: *which coordinates* carry the domain change? Stretch those axes.
- **Sample-level** answers: *which observations* look like the new category? Give them more CE mass.

Category-shift cold-start needs both questions. Feature-only helps when a few token dimensions dominate (`cnae9_text`, `amazon_cat`). Sample-only helps when novel support is spread but membership is informative (news20 boards). Product (`both`) is the default when unsure.

6 FULL-holdout endpoint (text; 3 seeds)

Dataset	uniform	feature_w	sample_w	both
news20_comp	0.378	0.386	**0.413**	0.389
news20_rec	0.458	0.457	0.473	**0.483**
news20_mix	0.283	0.281	**0.320**	0.303
cnae9_text	0.863	**0.889**	0.878	0.888
amazon_cat	0.692	0.705	0.695	**0.724**
mean overall acc	0.535	0.544	0.556	**0.557**
mean Δacc vs U	—	+0.009	+0.021	**+0.023**
mean novel OVA-AUC	0.768	0.764	0.768	0.762

Accuracy lift is stable; OVA-AUC stays roughly flat (ranking barely moves; decision-boundary / voting quality does).

Reproduce.

```
python3 scripts/run_text_full_weight_curves.py --multiseed
```

Figures: `results/text_full_weight_curves/figures/`.